



Tennessee Department of Transportation
Aeronautics Division
7335 Centennial Boulevard
Nashville, Tennessee 37209
615-741-3208

<https://www.tn.gov/tdot/aeronautics.html>

Overview:

The Tennessee Department of Transportation Aeronautics Division (Aeronautics) developed a framework that can be used by Aeronautics and its consultants to complete pavement rehabilitation alternatives. LCCA can be a valuable tool for comparing alternatives. The use of a LCCA methodology helps to shift the focus in project selection and toward those which have the lowest costs over the life of the pavement. The LCCA cycle cost estimates for TN airport pavement reconstruction projects to assist in the selection of the most cost-effective alternative. LCCA is not considered a complete Engineer's Estimate of Probable Costs.

Applied Pavement Technology, Inc. (APTech) has worked with Aeronautics to reconstruction alternatives to assist with evaluating the most cost-effective the guidelines provided in the Airfield Asphalt Pavement Technology Program study (AAPTTP Project 06-06) and Federal Aviation Administration (FAA) guidance of this release), *Airport Pavement Design and Evaluation*. The developed framework for future maintenance activities and their costs, intermediate rehabilitation activities, and of daily operation revenue associated with pavement closures to generate

Maintenance policies and timing have been based on analysis of data collected from the Management Program (PMP) and annual maintenance program. As pavement conditions deteriorate through the PMP, policies will be revisited. Pavement closure associated loss and cost alternatives. The analysis period has been set to 30 years based on review of

Depending on the durations maintenance activities, the framework has been limited facility uses or closures to implement repairs and rehabilitation.

Salvage value for the pavement alternatives is based on the assumed remaining value, the monetary calculation of the worth of the existing pavement. Adding value to recycling pavement materials at the end of their life, there is a suitability of the material for reuse, the future demand for the recycled materials that residual value is not used in these calculations.

Additional analysis inputs include discount rate and contingency fees (mobil established by Aeronautics and should only be revised after discussion with

If interested in the full report produced as part of developing this LCCA fram

tics) has sponsored the development of an analytical a reasonable life-cycle cost analysis (LCCA) of comparing alternatives of varying costs and service lives. away from alternatives with the lowest initial costs . This workbook assists in the development of life-cost with pavement type selection. It is not to be

develop this LCCA framework to compare feasible options. The developed LCCA framework considers n's *Life Cycle Cost Analysis for Airport Pavements* guidance in Advisory Circular 150/5320-6G (at the time framework considers construction (or initial) costs, activities and associated costs, salvage value and loss a Present Value for the alternatives.

ted as part of the Tennessee Statewide Pavement performance data continues to be collected ; of revenue is accounted for as indirect cost of d TN PMP and nationwide data.

n updated to account for lost revenue associated with

ining life as a percentage of construction cost. ent at the end of the service life, while certainly are enough uncertainties associated with the erial, and the costs associated with recycling the

ization and engineering). These have been Aeronautics.

ework, contact Aeronautics.



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Instructions:

This workbook has been set up to automate maintenance cost calculations and analyses for Aeronautics' review. This workbook contains privileged information by Aeronautics.

The use of the workbook is fairly straightforward, and the user should be able to generate a life-cycle cost estimate. Excluding this "Instructions" worksheet and "Overview" worksheets: "General Information," "Pay Items," "Maintenance Policies," and "Summary" worksheets are for reference only and should only be altered after discussion with Aeronautics. Additional worksheets for anticipated reconstruction alternatives are also included.

The following step-by-step procedures are provided to assist in using the workbook. Be sure to follow these directions in the logical, step-by-step fashion as this can write over any functions, commands, or equations. Make sure that functions work properly. There have also been issues with functions in the past. Opening the file in the main monitor should correct function issue, if they occur. If you see "CONTENT: The ActiveX content in this file is blocked", go to File > Options > Security > select "Prompt me before enabling all controls with minimal restrictions", click "Yes", and the content will be greyed out, ask your IT department to allow ActiveX for this file.

Note: Throughout the workbook, cells that require user-supplied inputs are in blue (and should not be modified by users).

1. Go to the "General Information" worksheet and enter in the relevant items (such as project airport, branch information, pavement area, and so on) that will be used for computations. Ensure the appropriate selection is made in cell D38.

2. Still in the "General Information" worksheet, click on the "Alternative Setup" button. A form will pop up for adding alternatives. The pop-up window includes the list of alternatives that will be used, update details and description, if desired, and click "Add" to add new alternatives. Alternatives can also be reorganized using the buttons "Move Up" and "Move Down" and a worksheet for each alternative will be created. A "Summary" worksheet will also be created.

3. Modify existing Pay Items, or insert new items in Table2 in the Pay_Items worksheet. The Pay_Items worksheet includes paving items only and does not include other items, such as electrical, drainage, etc.

4. To include lost airport revenue during runway closures, set cell D38 on the General Information worksheet to "Include Revenue". The average daily revenue for the selected airport is pulled automatically from the Airport Revenue worksheet, which covers the 17 airports meeting Aeronautics' activity criteria (cell D39 on the General Information worksheet reads "Missing Airport Revenue", lost revenue is carried as \$0, and a warning message is displayed to Aeronautics to add revenue data. To view the hidden worksheet: Home > Formulas > Show Hidden Worksheets.

5. Go to the worksheet for each alternative included in the LCCA and fill in cells for quantities (including accounting for indirect costs). Select the appropriate pay items in cells C13 through F10 (if needed; leave extra cells blank). Enter the quantities for each selected pay item. The quantities for pay items C13 through F10 are pre-filled from production-rate defaults (surface treatment 1 and slab work) and may be overridden with project-specific values.

6. Once the pay items and quantities have been filled in, the Net Present Worth will be calculated.

should be automatically filled in for the alternative. The "Summary" worksheet is updated once all individual alternative worksheets are completed. The Summary worksheet summarizes the costs for construction, maintenance, rehabilitation, lost revenue and salvage present in the alternative, the lowest-cost alternative and whether it still wins at the FAA AIP rate of 7%, and the impact of any changes. Use the "View Summary" button under Alternative Setup to open it.

If there are any issues or questions with the use of the LCCA workbook, please contact the LCCA team.

so as to expedite the overall preparation of life-cycle cost
tion and is intended for use only by consultants authorized

le to easily work their way through the development of a
ew" worksheet, the workbook initially contains four
l "Summary," The Pay Items and Maintenance Policies
n with Aeronautics. From the General Information
are created.

rkbook and in understanding how the series of worksheets
shion to avoid problems. Also, don't use the cut and paste
sure to "Enable Content" in Excel when the file is opened
running properly when multiple monitors are in use.
cur. If Excel shows a yellow bar reading "BLOCKED
Trust Center > Trust Center Settings > ActiveX Settings,
ck OK, then close and reopen Excel. If that setting is

ndicated in grey shading. Most other cells are calculated or

s in cells D9 through D39. This is general information
is needed to identify the project and complete the cost

ip" button at the bottom of the information list. A separate
t of available alternatives for this LCCA. Select which
k on "Add." The form can also be used to "Copy" or
s at the bottom of the form. Once completed, click "Close"
et is also populated as alternatives are completed.

worksheet as needed. Note that this LCCA is intended for
iage, and so on.

e "General Information" worksheet to Yes before creating
omatically from the hidden RevenueData worksheet,
shows the value). If the airport is not in that list, D39
g appears on each alternative worksheet; contact
rmat > Hide & Unhide > Unhide Sheet.

ells C4 through C8, and F4 through F10 (for analyses
through C22 for the alternative (only select the cells you
m in cells E13 through E22. Closure durations in cells F4
15,000 SY/day, mill and overlay 3,800 SY/day, PCC joint

orth (NPW) table and summary graph below the pay items

æet will also automatically fill with alternative costs as the
hows a results table for each alternative (initial
worth, net present worth, closure days), a line naming the
nd five charts. It updates by itself whenever an input
t and the buttons at the top of the Summary to return.

se contact Aeronautics.

**Airport Information:**

Airport Name:
Airport ID:
City:
Airport Owner:
State Region:
Design Consultant:
Project Number:
Project Name:
Date of LCCA:

Project Information:

Branch Type:
Branch Name:
Project Type:
Project Description:
Initial Construction Year:
Mainline Area, S.Y.:
Shoulder Area, S.Y.:
Markings, S.F.:
Markings, Type:

LCCA Parameters:

Analysis Period, years:
Discount Rate, percent:
Salvage Value, \$:
Mobilization, percent:
Engineering, percent:
Account Indirect Cost (Yes/No)
Airport Daily Revenue

Alternative Setup:**LCCA Summary:****Typical input values:**

N/A
N/A
N/A
N/A

--

Construction year; the analysis runs D33 year
Length x width / 9. TN GA runways: 4,000-6,0
Paved shoulders only; usually 0 at TN GA airp
Full runway marking set: roughly 2,000-3,000

	30	TDOT policy 30 years. FAA PGL 22-01 (2022,
	3	TDOT 3%. FAA now uses the OMB A-94 real
Remaining service life at end of analysis period (as percent of cost)	10	Mobilization 10% of pay items is the workbook
	5	Engineering 5% of pay items; now applied to i
	No	Yes counts lost airport revenue during closure

Alternative Setup

View Summary ▶

Typical Values ▶

Na
Ni

s from here.

00 ft x 75 or 100 ft = 33,000-6
ports.

SF per 1,000 ft; use the marki

) leaves the period to the engin
rate (2.0% in 2026); 7% was th

k default.

initial construction as well as M
s; 17 airports have revenue da

Division	Division Type	Pay Item No.
Part 3 – Sitework	Default	
	Sitework	P-101-5.1a
	Sitework	P-101-5.1b
	Sitework	P-101-5.2a
	Sitework	P-101-5.2b
	Sitework	P-101-5.2c
	Sitework	P-101-5.2d
	Sitework	P-101-5.3
	Sitework	P-101-5.4a
	Sitework	P-101-5.4b
	Sitework	P-101-5.5a
	Sitework	P-101-5.5b
	Sitework	P-101-5.5c
	Sitework	P-101-5.6
	Sitework	P-152-4.1
	Sitework	P-152-4.2
	Sitework	P-154-5.1
	Sitework	P-154-5.2
	Sitework	P-155-8.1
	Sitework	P-156-8.1
	Sitework	P-157-8.1a
	Sitework	P-157-8.1b
	Sitework	P-158-8.1
Part 4 – Base Courses	Base Course	P-207-5.1
	Base Course	P-208-5.1
	Base Course	P-208-5.2
	Base Course	P-209-5.1
	Base Course	P-209-5.2
	Base Course	P-219-5.1
	Base Course	P-219-5.2
	Base Course	P-220-6.1
Part 5 – Stabilized Base Courses	Stabilized Base Course	P-304-8.1
	Stabilized Base Course	P-306-8.1
	Stabilized Base Course	P-307-7.1
Part 6 – Flexible Pavements	Flexible Pavement	P-401-8.1a
	Flexible Pavement	P-401-8.1b
	Flexible Pavement	P-401-8.1c
	Flexible Pavement	P-403-8.1a
	Flexible Pavement	P-403-8.1b
	Flexible Pavement	P-403-8.1c
	Flexible Pavement	P-407-X.1
Part 7 – Rigid Pavement	Cement Concrete	P-501-8.1a
	Cement Concrete	P-501-8.1b
	Cement Concrete	P-501-8.1c
	Cement Concrete	P-501-8.1d
	Surface Treatment	P-608-8.1
Part 8 – Surface Treatments	Surface Treatment	P-623-8.1
	Surface Treatment	P-626-6.1
	Surface Treatment	P-629-8.1
	Surface Treatment	P-630-7.1a
	Surface Treatment	P-631-7.1b
	Surface Treatment	P-631-7.1b
Part 9 – Miscellaneous	Tack Coat	P-603-5.1
	Joint Sealant	P-605-5.1a
	Marking	P-620-5.1a
	Marking	P-620-5.1b

Grooving
Grooving

P-621-5.1A
P-621-5.1B

Pay Item Description

Pavement Removal, Asphalt
Pavement Removal, Concrete
Crack Repair (Type I), Asphalt
Crack Repair (Type II), Asphalt
Crack Seal, Concrete
Crack Repair, Concrete
Markings Removal
Spall Repair, Asphalt
Full-Depth Repair, Asphalt
Spall Repair, Concrete
Full-Depth Repair, Concrete
Slab Replacement, Concrete
Cold Milling
Unclassified Excavation
Embankment In-Place
Subbase Course
Separation Geotextile (P-154)
Lime Treated subgrade
Cement-Treated Subgrade
Cement Kiln Dust-Treated Subgrade
Lime Kiln Dust-Treated Subgrade
Fly Ash-Treated Subgrade
In-Place Full Depth Reclamation (FDR) Recycled Asphalt Aggregate Base Course
Aggregate Base Course
Separation Geotextile (P-208)
Crushed Aggregate Base Course
Separation Geotextile (P-209)
Recycled Concrete Aggregate Base Course
Separation Geotextile (P-219)
Cement-Treated Soil Base Course
Cement-Treated Base Course
Lean Concrete Base Course
Cement-Treated Permeable Base Course (CTPB)
Asphalt Surface Course
Asphalt Base Course
Asphalt Leveling Course
Asphalt Mixture Surface Course
Asphalt Mixture Base Course
Asphalt Mixture Leveling Course
Asphalt-Treated Permeable Base Course (ATPB)
Concrete Pavement, 5-inch
Concrete Pavement, 6-inch
Concrete Pavement, 9-inch
Concrete Pavement, 11-inch
Emulsified Asphalt Surface Treatment
Emulsified Asphalt Spray Seal Coat
Emulsified Asphalt Slurry Seal Surface Treatment
Thermoplastic Coal Tar Emulsion Surface Treatment
Refined Coal Tar Emulsion Without Additives, Slurry Seal Surface Treatment
Refined Coal Tar Emulsion With Additives, Slurry Seal Surface Treatment
Emulsified Asphalt Tack Coat
Joint Clean and Reseal, Concrete
Markings (prep, markings, reflective media)
Markings (prep, markings, non-reflective media)

Grooving, Asphalt
Grooving, Concrete

Units	Unit Cost	Average Pay Item Unit Cost		
		Middle	West	East
N/A	N/A			
S.Y.	\$4.20			Thickness adjustment or mill
S.Y.				Thickness adjustment or mill
L.F.	\$4.00			
L.F.	\$6.50			
L.F.	\$8.00			
L.F.	\$8.50			
S.F.	\$0.78			
S.F.	\$8.50			Thickness adjustment or mill
S.F.				Assumed Section: 4-inch H
S.F.	\$60.00			Thickness adjustment or mill
S.Y.	\$120.00			Thickness adjustment or mill
S.Y.	\$100.00			
S.Y.	\$5.00			Thickness adjustment or mill
C.Y.				
C.Y.				
C.Y.	\$55.00			
S.Y.				
S.Y.	\$10.00			Thickness adjustment or mill
S.Y.				Thickness adjustment or mill
S.Y.				Thickness adjustment or mill
S.Y.				Thickness adjustment or mill
S.Y.				Thickness adjustment or mill
S.Y.				Thickness adjustment or mill
C.Y.				
S.Y.				
C.Y.	\$65.00			
S.Y.				
C.Y.				
S.Y.				
S.Y.				Thickness adjustment or mill
C.Y.				If S.Y., then thickness adjust
C.Y.				If S.Y., then thickness adjust
C.Y.				If S.Y., then thickness adjust
TON	\$130.00			
TON	\$120.00			
TON	\$110.00			
TON				
TON				
TON				
TON				
S.Y.	\$79.00			Thickness adjustment or mill
S.Y.	\$82.00			Thickness adjustment or mill
S.Y.	\$85.00			Thickness adjustment or mill
S.Y.	\$91.00			Thickness adjustment or mill
S.Y.	\$4.00			
S.Y.	\$5.00			
S.Y.	\$7.00			
S.Y.				
S.Y.				
S.Y.				
GAL	\$4.50			Assume 0.1 gal/SY for mill ;
L.F.	\$7.00			Assume 12.5x12.5 slabs for
S.F.	\$0.67			
S.F.				

S.Y.
S.Y.

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and overlay
r these GA pavements



Table 1. New HMA maintenance policies.

Asphalt Maintenance
Maintenance 1
Maintenance 2
Maintenance 3
Maintenance 4
Rehabilitation 1
Maintenance 5
Maintenance 6
Salvage

Table 2. New PCC maintenance policies.

Concrete Maintenance
Maintenance 1
Rehabilitation 1
Salvage

Table 3. First HMA Overlay maintenance po

Asphalt Maintenance
Maintenance 1
Maintenance 2

Maintenance 3
Rehabilitation 2
Maintenance 4
Maintenance 5
Maintenance 6
Salvage

Table 4. PCC rehabilitation maintenance po

Concrete Maintenance
Maintenance 1
Rehabilitation 1
Salvage

Maintenance Item	Rate
Surface treatment, SYD	1.0000
Paint markings, SFT	1.0000
Surface treatment, SYD	1.0000
Paint markings, SFT	1.0000
Patching: 0.75% of area	0.0075
Crack sealing: 3.25% of area as linear feet	0.0325
Surface treatment: 100% of area	1.0000
Paint markings: 100% of markings	1.0000
Patching: 1.00% of area	0.0100
Crack sealing: 3.50% of area as linear feet	0.0350
Surface treatment: 100% of area	1.0000
Paint markings: 100% of markings	1.0000
Mill and overlay: 4-inch cold mill, tack coat, and asphalt overlay	1.0000
Pre-overlay crack repair: 1.5% of area as linear feet	0.0150
Pre-overlay patching: 0.5% of area	0.0050
Paint markings: 100% of markings	1.0000
Surface treatment: 100% of area	1.0000
Paint markings: 100% of markings	1.0000
Patching: 1.25% of area	0.0125
Crack sealing: 4.0% of area as linear feet	0.0400
Surface treatment: 100% of area	1.0000
Paint markings: 100% of markings	1.0000
Salvage value: 2 years (0.125) of mill and overlay	0.1250

Maintenance Item	Rate
Joint resealing: 100% of joints	1.0000
Crack sealing: 0.25% of area as linear feet	0.0025
Partial-depth patching: 0.13% of area	0.0013
Full-depth patching: 0.30% of area	0.0030
Slab replacements: 2% of area	0.0200
Full-depth patching: 0.30% of area	0.0030
Partial-depth patching: 0.13% of area	0.0013
Crack sealing: 0.30% of area as linear feet	0.0030
Joint sealing: 100% of joints	1.0000
Salvage value: 10 years (0.25) of initial construction	0.2500

licies.

Maintenance Item	Rate
Surface treatment: 100% of area	1.0000
Paint markings: 100% of markings	1.0000
Patching: 1.25% of area	0.0125
Crack sealing: 4.0% of area as linear feet	0.0400
Surface treatment: 100% of area	1.0000
Paint markings: 100% of markings	1.0000

Surface treatment, SYD	1.0000
Paint markings, SFT	1.0000
Mill and overlay: 4-inch cold mill, tack coat, and asphalt overlay	1.0000
Pre-overlay crack repair: 2% of area as linear feet	0.0150
Pre-overlay patching: 1% of area	0.0050
Paint markings: 100% of markings	1.0000
Surface treatment: 100% of area	1.0000
Paint markings: 100% of markings	1.0000
Patching: 1.25% of area	0.0125
Crack sealing: 4.0% of area as linear feet	0.0400
Surface treatment: 100% of area	1.0000
Paint markings: 100% of markings	1.0000
Surface treatment: 100% of area	1.0000
Paint markings: 100% of markings	1.0000
Salvage value: Need for reconstruction (zero salvage)	0.0000

licies.

Maintenance Item	Rate
Joint resealing: 100% of joints	1.0000
Crack sealing: 0.35% of area as linear feet	0.0035
Partial-depth patching: 0.2% of area	0.0020
Full-depth patching: 0.30% of area	0.0030
Slab replacements: 5% of area	0.0500
Full-depth patching: 0.30% of area	0.0030
Partial-depth patching: 0.15% of area	0.0015
Crack sealing: 0.30% of area as linear feet	0.0030
Joint sealing: 100% of joints	1.0000
Salvage value: Need for reconstruction (zero salvage)	0.0000

Year Applied
4
8
12
16
20
24
28
30

Year Applied
19
27
30

Year Applied
4
8

12
16
20
24
28
30

Year Applied
15
25
30

Alternative	Name	Initial Construction	Present Worth	Alternative Description
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LCCA SUMMARY

Everything from this column onward calculates from the alternative worksheets and updates by itself. Columns A-E and "Chart 1" are written by the Alternative Setup form as below.

Worksheet	Alternative	Type	Initial construction	Maintenance PW	Rehabilitation PW	Lost revenue PW	Salvage PW	Net present worth	vs. lowest NPW	Closure days in period	Runway availability

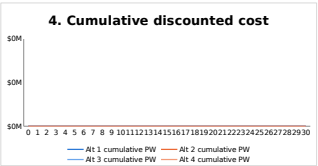
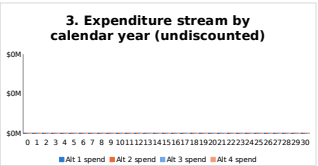
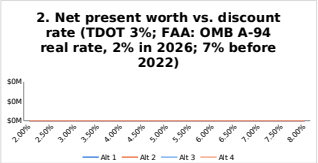
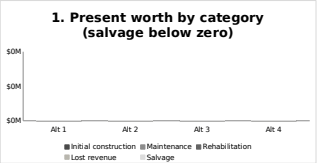
No alternatives yet. Go to General Information and click Alternative Setup.

COMPARISON (RealCost layout: agency cost, user cost, total; present worth and equivalent uniform annual cost)

Alternative	Agency cost PW	Agency EUAC	User cost PW (lost revenue)	User EUAC	Total PW	Total EUAC	vs. lowest (\$)	vs. lowest (%)	Lowest?

Agency cost = initial construction + maintenance + rehabilitation + salvage. User cost = lost airport revenue during runway closures, the airport-side counterpart of the user delay cost in the CaltransFTRisk RealCost layout. EUAC = $PW \times (1+i)^N / [(1+i)^N - 1]$ at the discount rate and analysis period on General Information. "vs. lowest (%)" is the difference as a share of the lowest total PW.

CHARTS



How to read: 1 shows where each alternative's cost sits; 2 whether the lowest-cost alternative holds at other discount rates (FAA now uses the OMB A-94 real rate, about 2%; 7% was the rule before 2022); 3 and 4 when the money is spent and when the higher first cost is paid back; 5 how often and how long the runway closes.

Reference sheet: nothing here feeds the calculation.

TYPICAL VALUES AND HELPERS FOR THE INPUTS

Reference only: nothing on this sheet feeds the calculation. Use it to sanity-check what you type on General Information and the alternative worksheets. Values are ranges from the cited sources as of the date in the source column; confirm against current bid tabulations for a specific project.

1. What TDOT Aeronautics manages (context for project size)						
Item	Value	Unit	Year			Source
Public-use airports in the Tennessee system (TASP 2040)		78 airports	2021			https://www.tasp2040.com/wp-content/uploads/2021/03/Updated-TASP-Inven
Commercial service / general aviation	6 / 72	airports	2021			https://www.tasp2040.com/wp-content/uploads/2021/03/Updated-TASP-Inven
Airports in the FAA NPIAS (eligible for AIP)		69 airports	2021			https://www.tasp2040.com/wp-content/uploads/2020/12/Chapter-7-Airport-
Airports in the TDOT pavement management network (APTech APMS, PCI per ASTM D5340)	~70	airports	2022			https://www.appliedpavement.com/tennessee-apms
PCI performance objective, runways / taxiways	78 / 75	PCI	2021			https://www.tn.gov/content/dam/tn/dot/aeronautics/TN%20Airport%20Mana
TASP system-wide estimate: pavement rehabilitation (10-yr life goal)		70,865,874 \$	2021			https://www.tasp2040.com/wp-content/uploads/2021/09/TASP-Cost-Estimate
TASP system-wide estimate: pavement new / reconstruction (20-yr life goal)		173,117,648 \$	2021			https://www.tasp2040.com/wp-content/uploads/2021/09/TASP-Cost-Estimate
State share on GA capital projects (federal or state funded)	up to 90% (10% local); 95/5 on federal discretionary		2023			https://www.tn.gov/content/dam/tn/dot/documents/170-02.pdf
Statewide Airfield Pavement and Marking Maintenance Program (crack seal, seal coat, markings)	100% state funded, no local share		2024			https://www.tn.gov/dot/aeronautics/apmm-contract.html
Pavement areas and PCI by airport (statewide tables)	see APTech IDEA portal		2022			https://idea.appliedpavement.com/hosting/tennessee/statewide-summary/i
NPIAS 2025-2029 classification of each TN airport (National / Regional / Local / Basic)	see Appendix A		2024			https://www.faa.gov/sites/faa.gov/files/airports/planning_capacity/npi

The workbook dropdown holds the TASP airports (count in the live section below). The 17 airports with revenue data are those meeting Aeronautics' activity criteria.

2. Typical runway geometry at Tennessee GA airports (for D26 mainline area and D28 markings)						
Airport	Runway	Length x width (ft)	Mainline area (SY)	Role	Note	Source
Murfreesboro Municipal (MBT)	18/36	4,753 x 100		52,811 GA, Middle	workbook MBT run used 52,778 SY	https://en.wikipedia.org/wiki/Murfreesboro_Municipal_Airport
Bomar Field, Shelbyville (SYI)	18/36	5,503 x 100		61,144 GA, Middle		https://en.wikipedia.org/wiki/Shelbyville_Municipal_Airport_(Tennessee)
Gatlinburg-Pigeon Forge (GKT)	10/28	5,506 x 75		45,883 GA, East	75-ft width	https://en.wikipedia.org/wiki/Gatlinburg%E2%80%93Pigeon_Forge_Airport
Outlaw Field, Clarksville (CKV)	17/35	5,999 x 100		66,656 GA, Middle	used in the new-project test	https://en.wikipedia.org/wiki/Clarksville%E2%80%93Montgomery_County_Re
McKellar-Sipes Regional (MKL)	2/20	6,005 x 150		100,083 GA, West	rebuilt 2024, ~\$28M all-in	https://www.wbbitv.com/2024/08/28/mckellar-sipes-regional-airport-set-
Smyrna (MQY)	1/19	5,546 x 100		61,622 GA, Middle	reconstructed 2025, ~\$13M all-in	https://www.smyrnaairport.com/news-events/smyrna-airport-runway-119-gr
Smyrna (MQY)	14/32	8,038 x 150		133,967 GA, Middle	former military	https://en.wikipedia.org/wiki/Smyrna_Airport_(Tennessee)
Typical TN GA runway		4,000 to 6,000 x 75 or 100	33,000 to 67,000		TASP Regional Service minimum width 100 ft	https://www.tasp2040.com/wp-content/uploads/2020/12/Chapter-7-Airport-
Markings area rule of thumb					2,000 to 3,000 SF per 1,000 ft of runway for a full runway marking set (centerline, thresholds, designators, aiming points, TDZ where applicable); use the marking plan quantity when available	engineering judgement; verify with the marking plan

Mainline area = length x width / 9. Shoulders (D27) are usually unpaved at TN GA airports; enter paved shoulders only. Area drives the closure-day defaults on the alternative sheets.

3. Unit costs: workbook defaults (Pay_Items, column F) against recent bids						
Pay item	Workbook default (live)	Unit	Recent bid range	Year	Reading	Source
P-401 Asphalt Surface Course		130.00 \$/ton	185 (low bid) to 232 (engineer's estimate)		2,025 Workbook default looks low against 2025 southeastern bids; confirm before relying on the default	https://aeronautics.sc.gov/sites/default/files/2025-05/250521_FLO_Reha
P-401 Asphalt Base Course		120.00 \$/ton	typically 5 to 10% below surface course			https://aeronautics.sc.gov/airport-development/sc-airport-capital-proj
P-501 Concrete Pavement, 9-inch		85.00 \$/SY	165 (8-in doweled apron, planning value)	2,023	Planning value from another state DOT; obtain a TN or southeastern PCC bid before use	https://www.dot.state.mn.us/aero/planning/documents/sasp/sasappendixf
P-209 Crushed Aggregate Base Course		65.00 \$/CY	not found in open sources; see SC bid tabs			https://aeronautics.sc.gov/airport-development/sc-airport-capital-proj
P-154 Subbase Course		55.00 \$/CY	not found in open sources; see SC bid tabs			https://aeronautics.sc.gov/airport-development/sc-airport-capital-proj
P-101 Cold Milling		5.00 \$/SY	4.70 (3-in, low bid) to 10.00 (variable depth, EE)	2,025		https://aeronautics.sc.gov/sites/default/files/2025-05/250521_FLO_Reha
P-608 Emulsified Asphalt Surface Treatment		4.00 \$/SY	TDOT APMM 2022 Summary of Bids	2,022	State maintenance contract prices apply to seal coats and crack sealing	https://www.tn.gov/dot/aeronautics/apmm-contract.html
P-620 Markings (reflective)		0.67 \$/SF	TDOT APMM 2022 Summary of Bids	2,022		https://www.tn.gov/dot/aeronautics/apmm-contract.html
All-in runway reconstruction, TN 2024-2025 (pavement + lighting + grading)		\$/SY	210 (Smyrna 1/19) to 280 (MKL 2/20)	2,025	Project-level figures; the LCCA compares pavement pay items only	https://www.smyrnaairport.com/news-events/smyrna-airport-runway-119-gr
State maintenance project example (Dyersburg): >200,000 LF crack seal, >121,000 SY seal coat, markings	820,000 \$			2,023	about \$5/SY of runway treated	https://www.stategazette.com/news/regional-airport-receives-state-gran

Pay_Items column F holds the default unit cost used by the alternative sheets (regional columns G-I are blank in v1.2.0). The defaults date from the 2022 framework; the 2025 comparison rows suggest the asphalt default is below current bids. Updating Pay_Items is the only place a cost needs to change.

4. Closure-day defaults built into the alternative sheets (cells F4:F10; override with project-specific values)						
Operation	Production rate in the formula	Days for a 6,000 x 100 ft runway (66,667 SY, 15,000 SF markings)	Where used	Note		Source
Surface treatment (seal coat)	15,000 SY/day		5 HMA Maintenance 1-6	each operation is rounded up to whole days		this workbook (MBT and SRB runs)

Markings: paint / layout and removal	20,000 SF/day and 50,000 SF/day		2 every HMA activity			this workbook
Patching	10,000 SY of pavement per day at the policy rate (0.75% to 1.25% of area)		1 HMA Maintenance 3, 4, 6			this workbook
Crack sealing	5,000 SY of pavement per day at the policy rate (3.25% to 4% of area)		1 HMA Maintenance 3, 4, 6			this workbook
Mill and overlay (4 in)	3,800 SY/day		19 HMA Rehabilitation 1	the largest HMA closure		this workbook
PCC joint resealing	about 1 LF of joint per SY of slab, 10,000 LF/day		7 PCC Maintenance 1 and Rehabilitation 1			this workbook
PCC crack sealing, partial- and full-depth patching	5,000 to 10,000 SY of pavement per day at the policy rate		3 PCC Maintenance 1			this workbook
PCC slab replacement (2% of area) + cure	1,000 SY/day plus 7 days cure		9 PCC Rehabilitation 1	the largest PCC closure		this workbook
Typical totals over 30 years		HMA 57 to 67 days in 7 closures; PCC 27 to 29 days in 2 closures	Summary column Q	MBT (52,778 SY) and CKV test (66,667 SY)		this workbook

Initial construction is not a closure in the model (the runway is assumed closed for construction regardless of alternative). Lost revenue = closure days x airport daily revenue x 100%; a per-category "share lost" factor is a policy decision listed in the change notes.

5. Economic parameters (D33 analysis period, D34 discount rate, salvage)

Parameter	Value	Applies to	Year		Reading	Source
TDOT policy discount rate (workbook default)		3.00 %			live from General Information D34	this workbook
TDOT policy analysis period (workbook default)		30.00 years			live from General Information D33	this workbook
FAA AIP: discount rate for pavement cost-effectiveness (PGL 22-01, June 2022)	OMB Circular A-94 Appendix C real rate for the analysis period	%		2022	Replaced the fixed 7% rule; the analysis period is now the engineer's choice (the fixed 20-year requirement was removed)	https://www.faa.gov/airports/aip/guidance_letters/aip_pgl_22_01
OMB A-94 Appendix C real rate, 20-year and 30-year (calendar 2026)		2 %		2026	Revised March 2026 (OMB memo M-26-09); 2025 values were 1.8% (10-yr) and 2.3% (30-yr)	https://www.whitehouse.gov/wp-content/uploads/2026/03/CircularA-94Appe
Legacy FAA AIP rule before PGL 22-01	7% and 20 years			2014	AIP Handbook (Order 5100.38D) pavement LCCA table; also the 1992 A-94 benefit-cost base case reinstated by OMB M-25-23 (April 2025) for BCA, not for cost-effectiveness	https://www.whitehouse.gov/wp-content/uploads/2025/02/M-25-23-Rescissi
FAA AC 150/5320-6E Appendix 1 (superseded)	4% and 20 years; straight-line residual value			2009	Dropped in 6F/6G; still widely cited	https://www.faa.gov/documentlibrary/media/Advisory_Circular/150_5320_6
AAPTP 06-06 airport LCCA guide (Auburn/NCAT)	4%, 20 years, NPW, remaining-life salvage; user cost includes lost airport revenue			2011	Basis for the "airport daily revenue" approach used here	https://www.eng.auburn.edu/research/centers/hcat/files/aatptp/Report.FI
Caltrans LCCA manual / RealCost	4% real; 20 / 35 / 55 years (CAPM / rehabilitation / new construction)			2013	Agency and user costs reported as present value and EUAC; the Summary comparison block follows that layout	https://dot.ca.gov/-/media/dot-media/programs/maintenance/documents/of
Structural design life (FAARFIELD, AC 150/5320-6G)		20 years		2021	design life, not service life	https://www.faa.gov/documentlibrary/media/Advisory_Circular/150-5320-6
Salvage: HMA		0.13 fraction of the last mill-and-overlay			live from Maintenance Policies D32 (policy: 2 of 16 years remaining)	this workbook
Salvage: PCC		0.25 fraction of initial construction			live from Maintenance Policies D46 (policy: 10 of 40 years remaining)	this workbook
Salvage method in FAA / AAPTP guidance	remaining life / expected life x cost of the last treatment (straight line)			2011	the workbook policy follows this form; the fractions above are the policy decision flagged in the change notes	https://www.eng.auburn.edu/research/centers/hcat/files/aatptp/Report.FI

The Summary sensitivity chart runs 2% to 8% so both the current FAA rate (about 2%) and the legacy 7% are on the curve. G10 on the Summary states whether the lowest-cost alternative changes at either rate.

6. Maintenance timing and service life (Maintenance Policies sheet against published experience)

Item	Workbook policy (live)	Published range	Year		Reading	Source
HMA surface treatment / markings	years 4, 8, 24	preventive-maintenance window PCI 60-80; inspect PCI every 3 years (AIP airports)		2014	FAA ACs give the PCI window, not fixed year intervals; 4-year cycles are the TDOT policy	https://www.faa.gov/documentlibrary/media/advisory_circular/150-5380-7
HMA patching + crack sealing + treatment	years 12, 16, 28	crack sealing, patching, seal coats per AC 150/5380-6C		2014		https://www.faa.gov/documentlibrary/media/advisory_circular/150-5380-6
HMA mill and overlay	year 20	asphalt runways reach PCI 70 in about 12 to 15 years (FAA 2014 performance trends)		2014	policy year 20 is at the optimistic end of the published range	https://www.acpa.org/wp-content/uploads/2025/03/ACPA-Achieving-the-Goa
PCC joint resealing + patching	year 19	hot-pour sealant 3-8 yr; silicone 8-15 yr; compression seals 15-20 yr			policy year 19 assumes silicone-class sealant life	https://www.tarmacview.com/glossary/joint-sealant/
PCC slab replacement (2%) + joints	year 27	PCC runways reach PCI 70 at about 40 years (FAA 2014)		2014		https://www.acpa.org/wp-content/uploads/2025/03/ACPA-Achieving-the-Goa
Expected service life	HMA 20 yr design; PCC 30-40 yr	FAA Extended Airport Pavement Life program targets 40 years		2024		https://www.airporttech.tc.faa.gov/Portals/0/FactSheets/Extended%20Air
Treatment fact sheets with life and cost by treatment		ACRP Report 159 (GA airports) and ACRP Synthesis 22		2016	use for project-specific overrides of the policy years	https://nap.nationalacademies.org/catalog/23649/
Minimum layer thickness, AC 150/5320-6G		P-401 surface 4 in (>= 12,500 lb); P-209 base 6 in; P-501 6 in (5 in under 12,500 lb)		2021	check the pay-item quantities against these minimums	https://www.faa.gov/documentlibrary/media/Advisory_Circular/150-5320-6

Maintenance Policies drives every activity year and rate; this section only shows them next to the published ranges. Sources were read from search excerpts in September 2026 because the FAA and OMB sites could not be opened from the build environment; confirm the cited paragraphs before quoting them to FAA.

Airport daily revenue used for lost-revenue cost (the 17 airports with revenue data; live from RevenueData)

Airport ID	Daily revenue (\$/day)	Lost revenue, 7-day closure (\$)	Lost revenue, 19-day closure (\$)	Columns holding text (counted as \$0)	Source
MKL	8,505.94	59541.58	161612.86	other	RevenueData sheet in this workbook (hidden)
SRB	4,344.12	30408.84	82538.28	other	RevenueData sheet in this workbook (hidden)
MBT	8,841.38	61889.66	167986.22	other	RevenueData sheet in this workbook (hidden)
NQA	6,291.00	44037	119529		RevenueData sheet in this workbook (hidden)
DKX	6,703.00	46921	127357		RevenueData sheet in this workbook (hidden)
SYI	30,816.64	215716.48	585516.16		RevenueData sheet in this workbook (hidden)
GCY	1,800.00	12600	34200	tenant	RevenueData sheet in this workbook (hidden)
THA	939.26	6574.82	17845.94	tenant	RevenueData sheet in this workbook (hidden)
GKT	5,441.17	38088.19	103382.23	tenant other	RevenueData sheet in this workbook (hidden)
JWN	89,240.00	624680	1695560	tenant	RevenueData sheet in this workbook (hidden)
MOR	8,801.40	61609.8	167226.6	tenant other	RevenueData sheet in this workbook (hidden)
MQY	8,841.38	61889.66	167986.22	other	RevenueData sheet in this workbook (hidden)
RZR	4,100.00	28700	77900		RevenueData sheet in this workbook (hidden)
XNX	81.97	573.79	1557.43	tenant other	RevenueData sheet in this workbook (hidden)
M91	2,058.00	14406	39102		RevenueData sheet in this workbook (hidden)
CKV	3,239.39	22675.73	61548.41	tenant other	RevenueData sheet in this workbook (hidden)
M54	1,027.81	7194.67	19528.39	100LL Jet-A	RevenueData sheet in this workbook (hidden)

Same SUMIF chain as General Information D39: text entries in RevenueData (" \$- ", " NA ", descriptions on XNX and M54) count as zero. 7 days is a typical surface-treatment closure and 19 days a mill-and-overlay closure for a 6,000 x 100 ft runway (see production rates).

Airports in the General Information dropdown					
Item	Value				Source
Airports selectable in D9	79.00				General Information, hidden columns L:P
Airports with revenue data	17.00				RevenueData
Airports with a Middle / West / East region tag	33 / 21 / 25				General Information, hidden columns L:P